The Design of Providing Location Services in Metropolitan with 3D Ellipse Propagation Model within a Cellular Radio Network

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Outline

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- 2. Related Work
- 3. 3D Ellipse Propagation Model
- 4. Future work
- 5. Q & A

1. Background

Positioning Technology

Global Positioning System (GPS)

- A GPS receiver measures the distances from 3 or more satellites and utilizes a mathematical method, called “Triangulation” to calculate the actual position

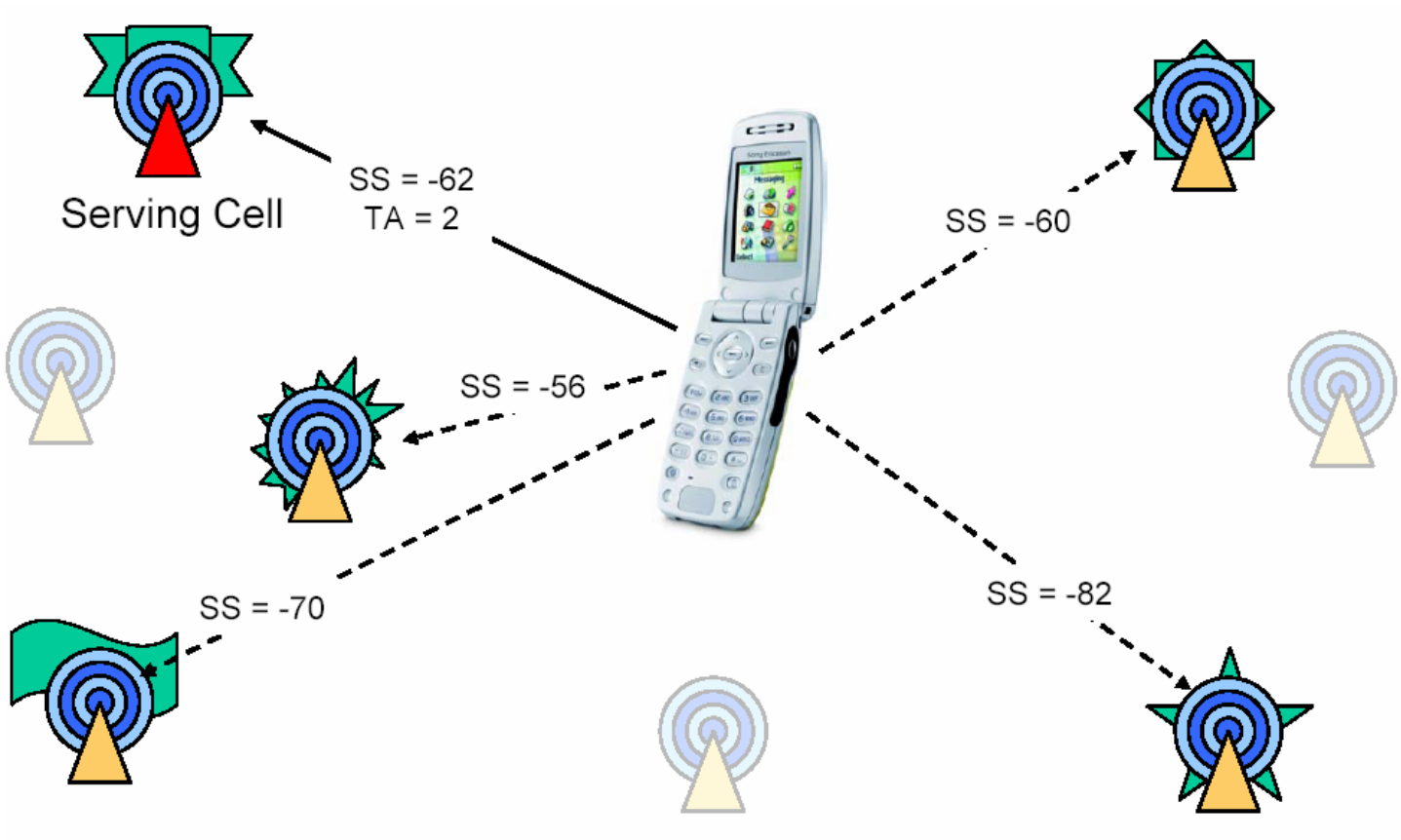
- Blind Spot Problem: Signals from the satellites would be blocked, and reflected by tall buildings.

Positioning Technology

Location Estimation Algorithms based on a Cellular Network

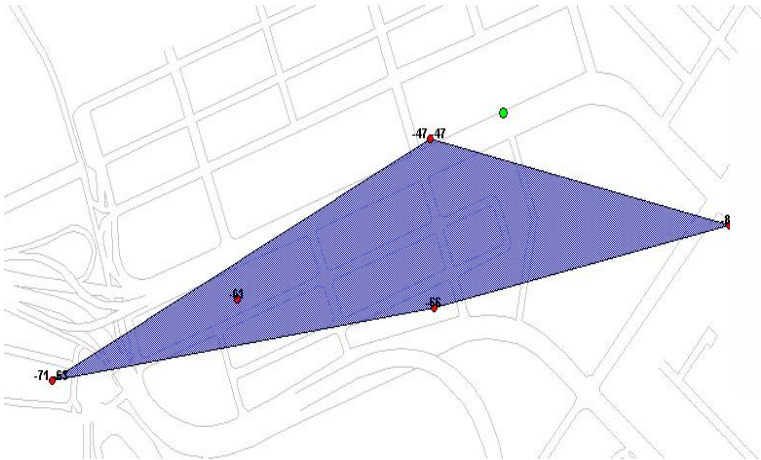
- CDMA: ToA, TDoA, E-OTD, AoA
- GSM: Signal strength based Algorithms
- Feasible for Metropolitan areas like Hong Kong
- Only the signal strength based algorithms can be applicable to all cellular networks

GSM Network



2. Related Work

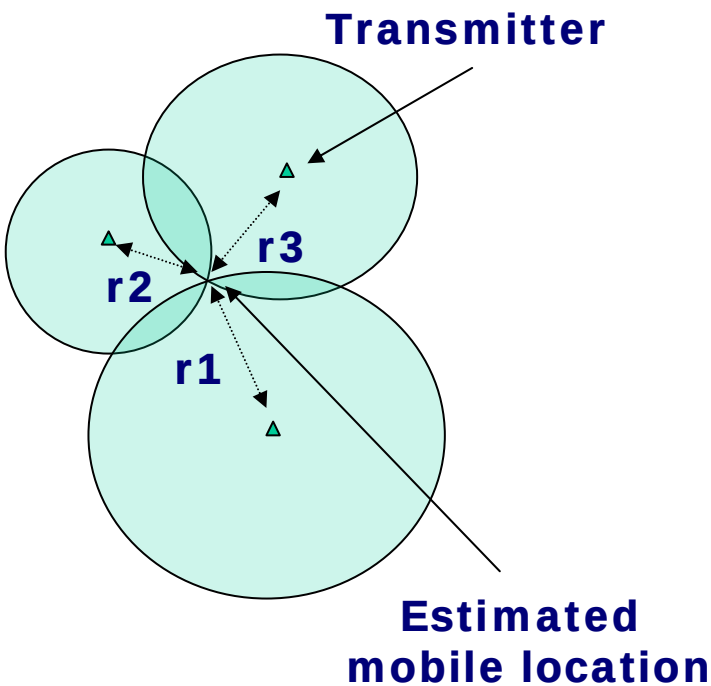
Center of Gravity Algorithm (CG)



$$\begin{cases} x = \dfrac{x_1 s_1^{-\alpha} + x_2 s_2^{-\alpha} + x_3 s_3^{-\alpha} + \ldots + x_n s_n^{-\alpha}}{s_1^{-\alpha} + s_2^{-\alpha} + s_3^{-\alpha} + \ldots + s_n^{-\alpha}} \\ y = \dfrac{y_1 s_1^{-\alpha} + y_2 s_2^{-\alpha} + y_3 s_3^{-\alpha} + \ldots + y_n s_n^{-\alpha}}{s_1^{-\alpha} + s_2^{-\alpha} + s_3^{-\alpha} + \ldots + s_n^{-\alpha}} \end{cases}$$

The “CG” estimation is always within the convex hull regardless the actual position of the MS is inside or outside this convex hull area

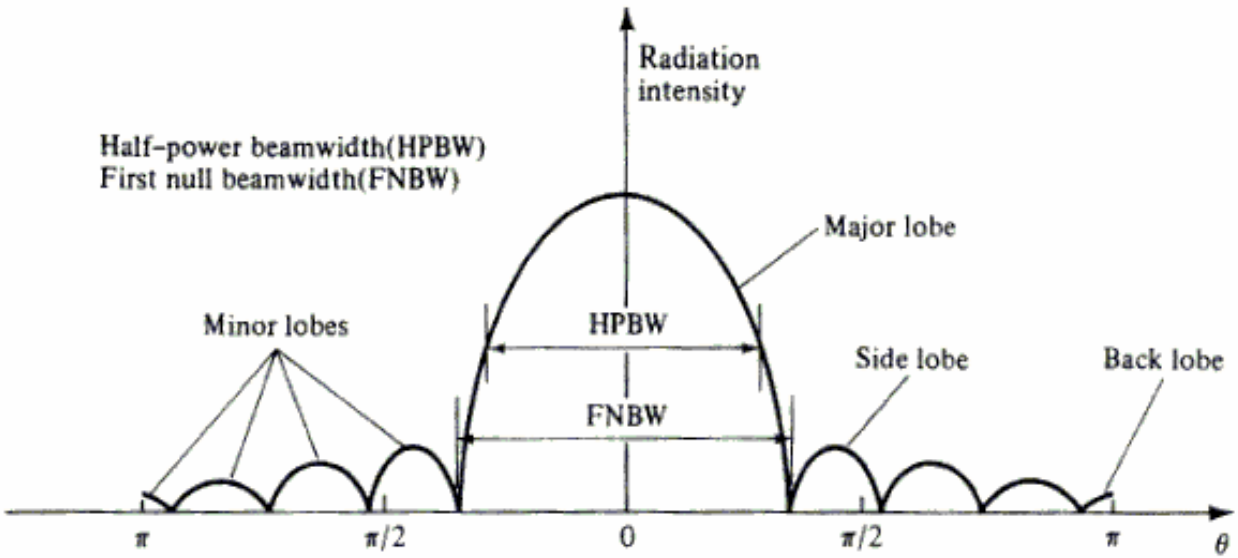
Circular Trilateration Algorithm (CT)



$$\begin{cases} (x-x_1)^2+(y-y_1)^2=(\frac{k}{s_1{}^{\alpha}})^2 \\ (x-x_2)^2+(y-y_2)^2=(\frac{k}{s_2{}^{\alpha}})^2 \\ (x-x_3)^2+(y-y_3)^2=(\frac{k}{s_3{}^{\alpha}})^2 \end{cases}$$

These three circles may not always intersect in the **one** point

The Directional Antenna



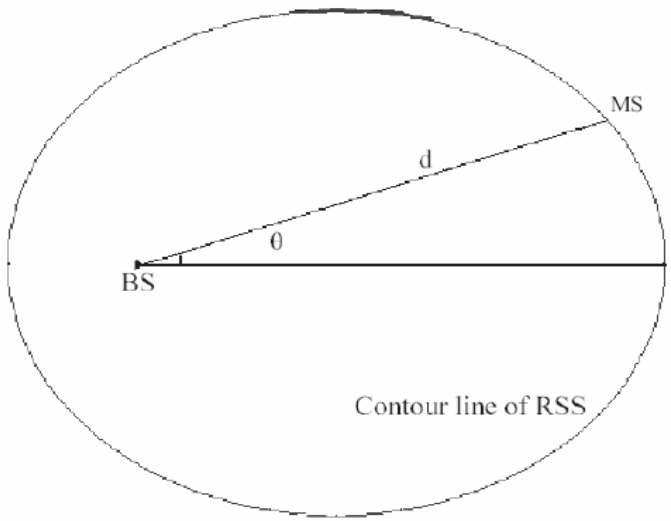
The Ellipse Propagation Model

Consider the contour line of the signal strength as an ellipse

$$d = k(s_0/s)^{1/\alpha}(1-e)/(1-e\cos(\theta))$$

where

d is the distance between the MS and the BS;
k is the proportion constant;
s_0 is the transmitting power of the BS;
s is the signal power received;
e is the eccentricity value of the ellipse, it describes the figure of the contour line of the signal strength ;
θ is the deviation between the ellipse principal axis and the line of the MS and the BS;
α is called the path loss exponent.



Two Algorithms based on EPM

- The Geometric Algorithm with EPM
- The Iterative Algorithm with EPM

3. 3D Ellipse Propagation Model

- Extend Ellipse Propagation Model into 3D space.
- Use Four BSs to provide the 3D location of MS.

4. Future work

- Sample Data
- Train Model
- Application

5. Q & A

Thank you!